SANDEEP TAYAL

Python | FastAPI | AI Backend | SQL

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PROFILE

Bachelors in Computer Science student focused on AI Backend Engineering and Machine Learning applications. Building backend applications using FastAPI, integrating machine learning models and RAG-based AI systems with SQL databases, authentication, and Docker.

EDUCATION

DN College, Meerut — Bachelors in Computer Science

Academic Performance: 10th — 92.4% | 12th — 90.6%

PROJECTS

Company AI Knowledge Assistant

FastAPI | LangChain | RAG | Mistral AI Embeddings | Chroma | Sarvam AI | MySQL | SQLAlchemy | JWT | Streamlit | Docker | Docker Compose

- Built a RAG-based AI chatbot that answers questions from company documents using Retrieval-Augmented Generation.
- Developed a FastAPI backend with a LangChain-based RAG pipeline for document retrieval and context-aware responses.
- Used Mistral AI embeddings with Chroma for storing and retrieving relevant document information.
- Integrated Sarvam AI as the LLM and used MySQL + SQLAlchemy for user and conversation-history storage with JWT authentication.
- Built a Streamlit frontend and containerized the application using Docker and Docker Compose.

GitHub: Repository | **Docker Hub:** Repository

Insurance Risk & Cost Predictor

FastAPI | Scikit-learn | SQLAlchemy | MySQL | Docker | AWS EC2

- Built an ML-based application for insurance risk and cost prediction using a trained Scikit-learn model.
- Developed REST APIs with FastAPI, including prediction and authentication functionality.
- Implemented Pydantic validation and database operations using SQLAlchemy + MySQL.
- Added derived features including BMI, lifestyle score, age group and city tier for prediction.
- Implemented JWT authentication and Swagger/OpenAPI documentation.
- Containerized the application using Docker and worked with AWS EC2 for deployment.

GitHub: Repository | **Docker Hub:** Repository

Loan Risk Assessment System

FastAPI | Scikit-learn | SQLAlchemy | MySQL | JWT | Pydantic

- Built an end-to-end ML-based loan risk assessment system with a FastAPI REST API.
- Integrated a trained Scikit-learn model for loan risk prediction.
- Used MySQL and SQLAlchemy ORM for persistent data storage.
- Implemented JWT authentication and Pydantic-based request validation.
- Added environment-based configuration and Swagger UI/OpenAPI documentation.
- Built the application with a backend-focused architecture suitable for deployment.

GitHub: Repository | **Docker Hub:** Repository

TECHNICAL SKILLS

Languages: Python, C, SQL

Backend: FastAPI, REST APIs, Pydantic, SQLAlchemy, JWT

Database: MySQL

ML & Data: Scikit-learn, Pandas, NumPy

AI / RAG: RAG, LangChain, Chroma, Mistral AI Embeddings, Sarvam AI

Tools & Cloud: Git, GitHub, Docker, Docker Compose, AWS EC2, VS Code

Frontend: Streamlit